Diseases of Colocasia

1. Phytophthora blight - *Phytophthora colocasiae*
2. Pythium root and corm rot - *Pythium aphanidermtum*, *P. ultimum*, *P. myriotylum*, *P. irregulare* and *P. gracile*

Phytophthora blight - *Phytophthora colocasiae*

Small, 3 to 4 mm, brown spots on an upper leaf surface surrounded by light green halos are often the first symptoms of disease

Enlarged spots

Phytophthora blight - *Phytophthora colocasiae*

Chlorotic halos around leaf spots, with shot-hole lesions scattered

Phytophthora blight - *Phytophthora colocasiae*

Expanding taro leaf blight lesion with distinct, circular zones.

Sporangia of Phytophthora colocasiae forming a whitish ring at the expanding margin of a lesion.

Phytophthora blight - *Phytophthora colocasiae*

Reddish orange exudate produced by an infected plant.

Dry weather causes typical water-soaked taro leaf blight lesions to become dried and brownish gray. The droplets of hardened plant exudate.

Phytophthora blight - *Phytophthora colocasiae*

The photosynthetic capacity of this leaf reduce by 75%, under wet conditions leaf will collapse within 24 hours.

Two dead taro leaf blades hanging from their petioles. Under wet conditions leaves and petioles collapse.

Phytophthora blight - *Phytophthora colocasiae*

Symptoms

- The disease appears in the form of **small dark, round spots** on the leaf
- The spots rapidly enlarges, coalesce followed by **drying and death of the entire leaf.**
- Later the spots become dry and affected portions drop out giving a **shot hole appearance.**
- **The spots are with margins of brown, green and yellow shades.**
- Often, drops of **yellow liquid ooze out** from the affected areas.

Phytophthora blight - *Phytophthora colocasiae*

Symptoms contd…

- **Infection on petiole extends downwards infecting stem and corms**
- Leaves collapse, from **the fallen leaves the pathogen spreads to corms**
- **Corms may rot in the field under wet conditions,** but rot spreads **more rapidly in the storage - rotted corms emits foul smell**

Disease cycle:

- Oospores in soil/ Sporangia and zoospores

Phytophthora blight - *Phytophthora colocasiae*

Management

- Crop rotation
- Field sanitation
- Disease free seed
- Avoid close planting
- Provide good drainage
- Resistant varieties; **Muktakeshi, Jhankri, Akina, poonam, sakir-II simla, bhadia, pusa**
- **Early planting (Feb – March)**
- Spraying with metalaxyl or copper fungicides

Pythium root and corm rot - *Pythium aphanidermtum, P. ultimum, P. myriotylum, P. irregulare and P. gracile*

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Pythium root and corm rot –*Pythium aphanidermtum, P. ultimum, P. myriotylum, P. irregulare and P. gracile*

Symptoms

Root rot

- **Base of the corm usually rot**, light yellow to grey blue / dark purple
- Diseased corms assume **parallel sided bottle neck shape**
- **Healthy root oblong / ovoid**
- **Roots decay- rhizomes soft, cheese like**
- **Petiole base become water soaked – leaves become chlorotic-** collapse

Pythium root and corm rot –*Pythium aphanidermtum, P. ultimum P. myriotylum, P. irregulare and P. gracile,*

Symptoms contd….

Corm rot

- **Corms - soft, exude a watery substance**
- **Deficient in starch**
- **Suckers of Pythium rotted plants** liable to develop disease symptoms

Disease cycle:

Oospores in soil/ Sporangia and zoospores

Pythium root and corm rot - *Pythium aphanidermtum, P. ultimum, P. myriotylum, P. irregulare and P. gracile*

Management:

- Crop rotation
- Ploughing and proper drying of the field
- Field sanitation
- Disease free seed
- Avoid close planting
- Provide good drainage
- Drenching- metalaxyl or copper fungicides

Dasheen mosaic (*Dasheen Mosaic Virus*)

feathery mottle

vein necrosis

leaf roll

severe
leaf deformity

Symptoms:

Feathery mottle, Vein necrosis

Leaf roll, Leaf deformity

Vectors

Myzus persicae,
Aphis craccivora

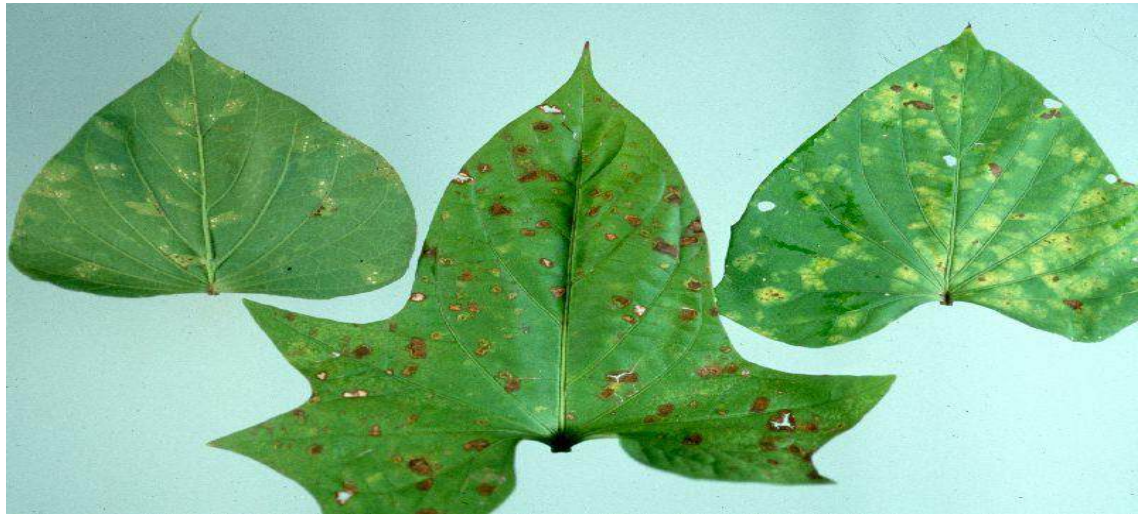
Transmitted in a non-persistent manner

Diseases of Sweet potato

1. White rust - *Albugo ipomoea*
2. Fusarial wilt, stem rot, surface rot - *Fusarium oxysporum* f. sp. *batatas*
3. Soil rot, soft rot or pox - *Streptomyces ipomoea*

White rust - *Albugo ipomoea*

White rust - *Albugo ipomoea*



White rust - *Albugo ipomoea*

Symptoms

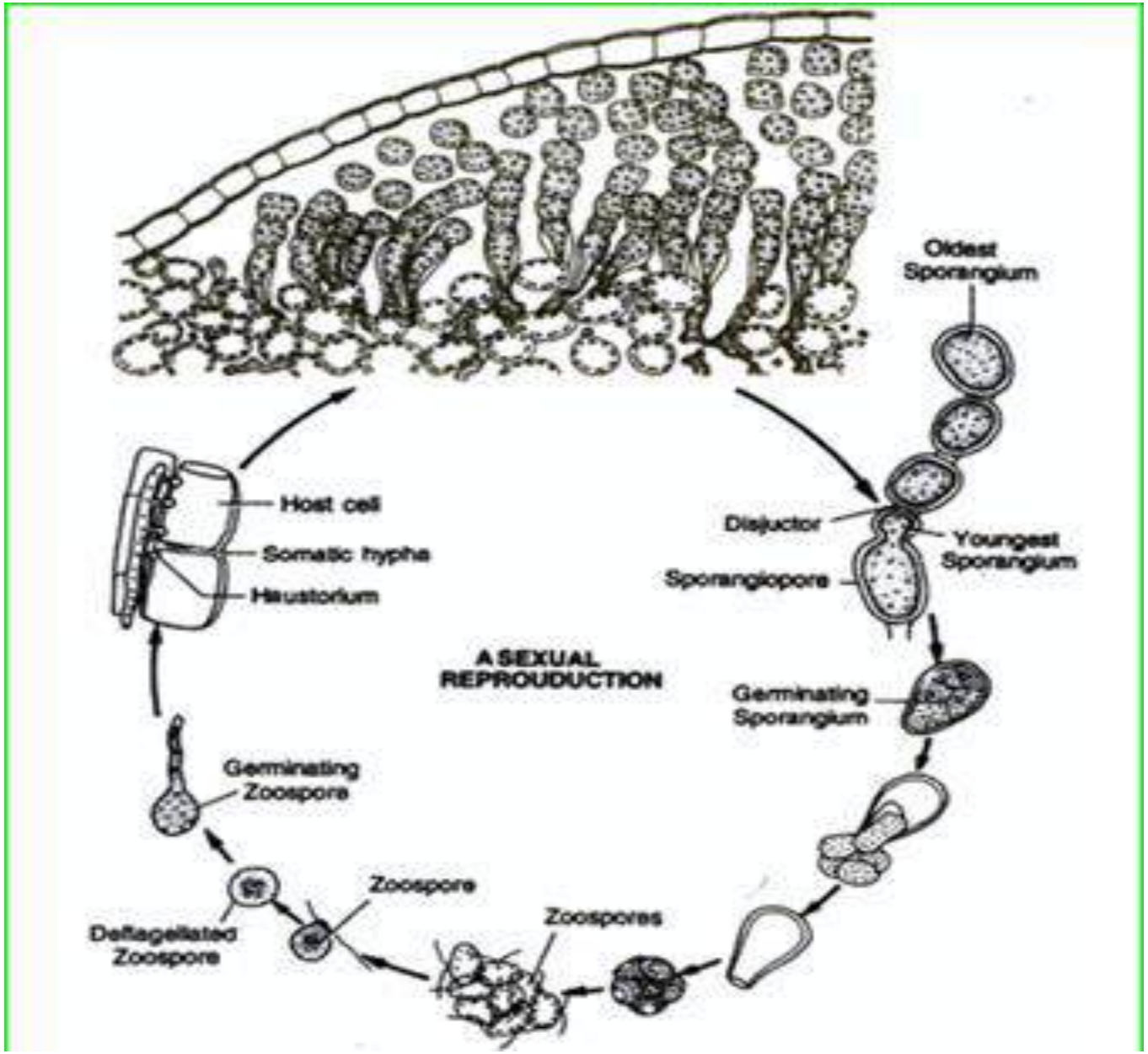
- Disease appears **late stage in plants with optimum growth**
- **Affects the older leaves** near the centre of the hill
- The most obvious symptom on the upper surface of leaves as chlorotic or yellowish blotches, initially roundish to angular where they are limited by veins.
- On the lower surface, small pustules develop which later open and expose whitish masses of sporangial pustules.

White rust - *Albugo ipomoea*

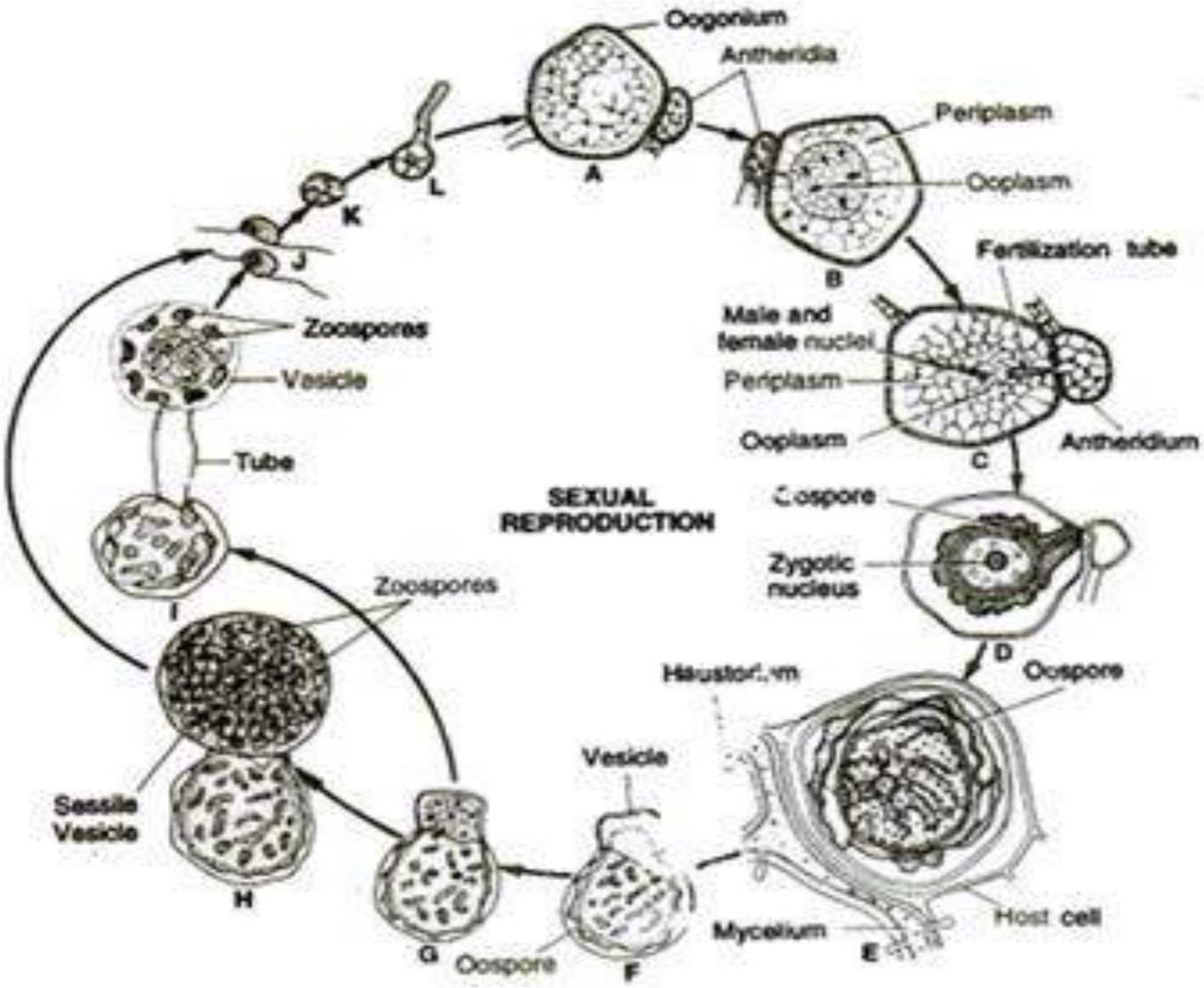
Symptoms contd…

- After sporulation, the infected tissue dies, forming irregular-shaped brown lesions
- **In some cultivars depending on the growing conditions,** infection induces the development of **galls of raised, thickened tissue.**
- Galls may develop **on leaves, petioles, stems and flowers.**
- When pustules erupt, the galls become covered in the white sporangia.
- Diseased plants show **general distortion, defoliation and flower abortion.**

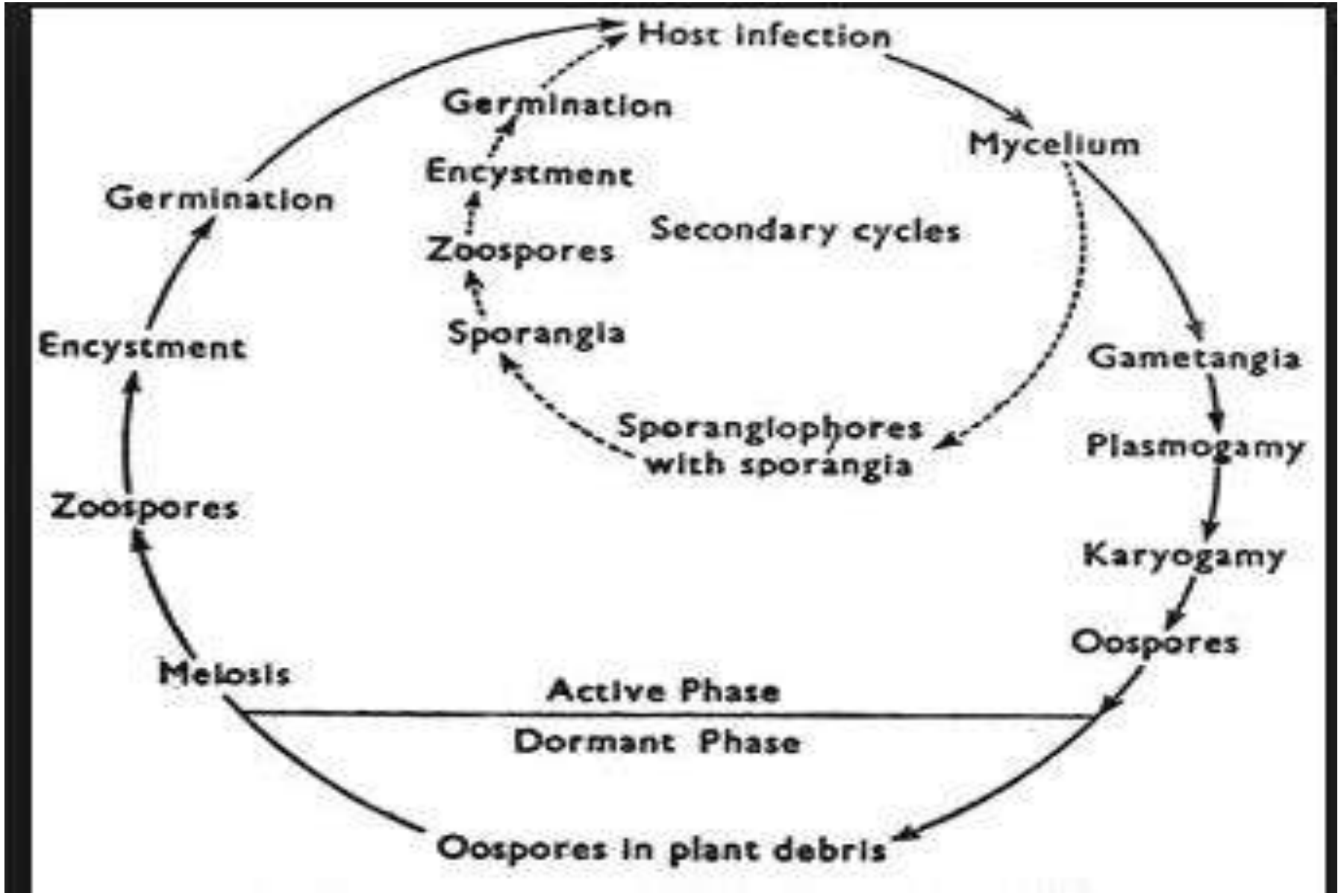
White rust - *Albugo* sp



White rust - *Albugo* sp



White rust - *Albugo* sp



White rust - *Albugo ipomoea*

Disease cycle

PSI: Infected plant debris/ oospores

SSI: Zoospores

Management

- Sanitation,
- Use of free setts
- Crop rotation

Fusarial wilt, stem rot, surface rot - *Fusarium oxysporum* f. sp. *batatas*

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Fusarial wilt, stem rot, surface rot - *Fusarium oxysporum* f. sp. *batatas*

Distribution and importance

- This disease is found in most of the sweet potato cultivated areas and is **more important in temperate areas than in the tropics.**

Symptoms

- Initial symptoms on sweet potato are **interveinal yellowing of the leaves followed by complete yellowing of the lower**, older leaves during the rapid growth stage.
- Leaves later wilt and abscise, stunting results and eventually death of the plant.

Fusarial wilt, stem rot, surface rot - *Fusarium oxysporum* f. sp. *batatas*

Symptoms contd..

- **Necrosis of the stem vascular bundles** occurs with brown to purple discoloration
- The **vines may turn tan to light brown**.
- Diseased plants may **manage to produce storage roots,** but these usually have some **discoloured infected vascular tissues**.
- **Rot may advance in storage** or the disease may be transmitted by infected cuttings to the next crop.
- Dying vines often have a pinkish fungal growth with numerous macroconidia and microconidia of *F. oxysporum* f.sp. *batatas*

Fusarium surface rot- *Fusarium oxysporum, F. solani*

Fusarium surface rot- *Fusarium oxysporum, F. solani*

Distinct from wilt pathogen - Most common disease
More losses than any other storage diseases

- Surface rot is **prevalent when harvested mechanically**
- **Soil is dry before harvest** (causing increased skinning of sweet potatoes) **wet and cold at harvest**
- After digging - **before curing** – if tubers exposed to high / low temperatures for extended periods **/ wounded tissue dessicates**
- Occasionally - **before harvest - roots injured mechanically**/ growth cracks/ nematodes/ insects and soil pests

Fusarium surface rot- *Fusarium oxysporum, F. solani*

Symptoms

- Some strains - spread - **mother roots to sprouts causing stem canker – necrosis - distort**
- Initial lesions are circular, tissues near the margins -**orange to light brown, spongy and moist**
- Older lesions dark brown, **spongy and dry characterized by lens shaped cavities**
- Decay remains shallow, usually not extending beyond vascular ring
- Cavities with fungal growth on the inner surface
- When **infected tubers - stored for longer periods, the tissue around the lesions dries and becomes shrunken**, and the tubers eventually becomes hard and mummified.

Fusarial wilt, stem rot, surface rot - *Fusarium* spp

Biology

- The fungus is soil-borne and specific to sweet potato and a few close relatives, barley and flue-cured tobacco.
- It can survive in the soil and in debris for several years.
- **Though terminal cuttings are usually pathogen free**, roots and cuttings from the base of the vine can be infected.

Fusarial wilt, stem rot, surface rot - *Fusarium oxysporum* f. sp. *batatas*

Biology contd….

- Movement of infested soil on tools and by animals can lead to outbreaks in new areas.
- The disease occurs under a variety of environmental conditions
- Yield reduction depends on the stage of plant growth when disease occurs.

Disease cycle

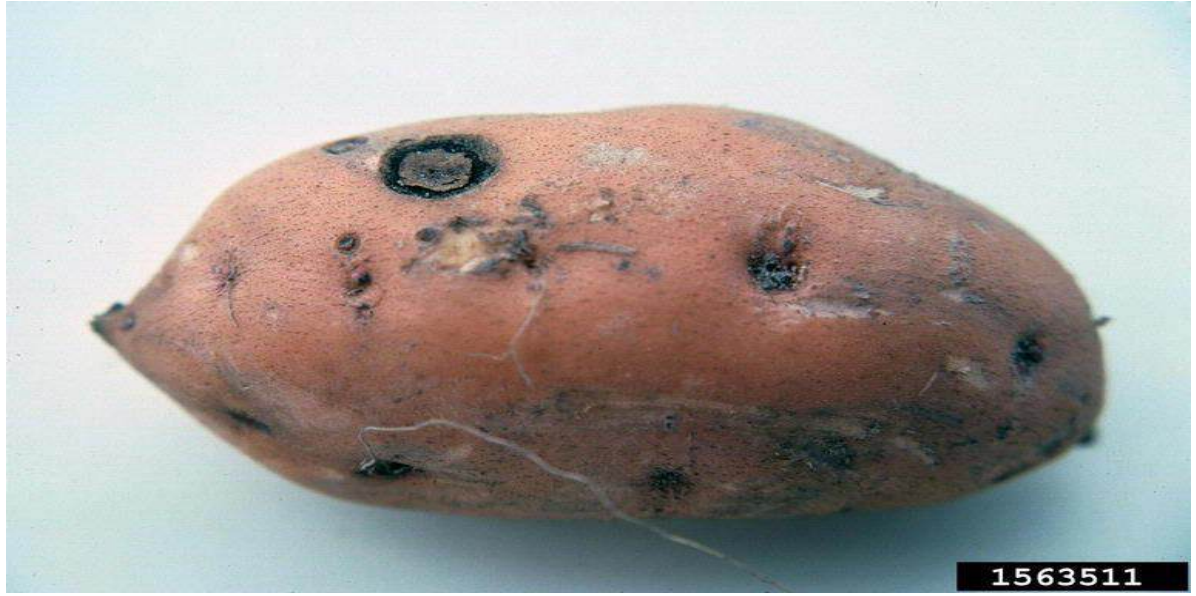
- **As chlamydospores in plant debris, soil**

Fusarial wilt, stem rot, surface rot - *Fusarium oxysporum* f. sp. *batatas*

Management

- Use disease free seed
- Treat the seed tubers
- Good sanitation will help reduce the impact of the disease and limit its spread.
- Varietal resistance and by breeding in some countries resistant varieties are developed and released.
- **Minimize injuries at the time of harvest** and handling
- **Harvest when soil is dry, treat and cure** immediately

Soil rot, soft rot or pox - *Streptomyces ipomoea*



Soil rot, soft rot or pox - *Streptomyces ipomoea*

Distribution and importance

- This disease reduces yield and can be destructive in **some areas of the United States and Japan.**

Symptoms

- The first indication of the disease is **an extensive chlorosis and bronzing of the foliage as due to the destruction of fibrous roots**
- Plants with small, **pale leaves, stunted stems**
- Reduction in number of **feeder roots**
- **On under ground stem** – small, dark, elongated lesions

Soil rot, soft rot or pox - *Streptomyces ipomoea*

Symptoms contd..

- **Fleshy tubers** -initially water soaked lesions on the surface- **later develop as pits**
- On storage roots, besides dark **brown corky necrotic lesions, may be with cracks** radiating from the center
- **Tubers may be girdled, but continue to grow on either side of the constricted area** and malformed resemble **dumbbell**
- **Dry and poor soil** favors disease development

Soil rot, soft rot or pox - *Streptomyces ipomoea*

Biology

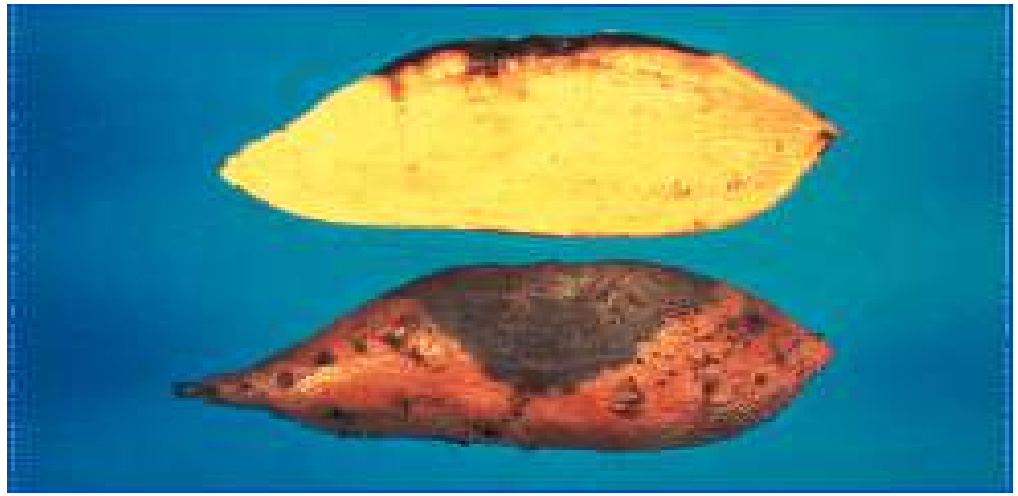
- Soil rot causes more damage in dry alkaline soils. The pathogen can **survive in soil for long periods.**

Management

- Curing of roots immediately after harvest at 85 to 95° F, and 85 to 90 % relative humidity / 5 to 10 days.
- **Collect seed tubers** from areas where the disease is not present.
- **Maintaining soil moisture** helps reduce disease incidence.
- Using **sulfur to reduce soil pH to 5** is another alternative, but large amounts of this element are required.

Black Rot - *Ceratocystis fimbriata*

Roots with black rot lesions (left and right) and uninfected roots (center).



Internal and external views of roots infected with black rot

Black Rot - *Ceratocystis fimbriata*

- **Small, circular, slightly sunken, dark brown spots** are the initial symptoms of black rot
- Spots enlarge and appear **greenish black to black when wet and grayish black when dry.**
- Within the spots are small, black fungal structures (perithecia) with long necks which appear to the naked eye as dark bristles.
- The rot usually remains **firm and shallow.**
- If secondary fungi or bacteria invade the tissue however, the flesh beneath the spot turns black, and this blackened area may extend to the center of the root
- Tissue near the discolored area may have a bitter taste. Eventually, the entire root may rot.
- **Roots may appear healthy at harvest but rot in storage, during transit, or in the market.**

Disease cycle: Soil borne

Black Rot - *Ceratocystis fimbriata*

Management

- Control black rot with crop rotation, since most crops are unaffected by the disease.
- Disinfect seedbeds if a clean site is unavailable.
- Propagate plants from healthy stem cuttings.
- Cure roots immediately after harvest.
- Apply a postharvest fungicide.
- Do not wash and package roots showing symptoms of black rot.
- Decontaminate equipment that comes into contact with an infected crop.
- **Spray empty washing machines and crates with a fungicide.**
- **Fumigate storage structures.**